디지털 논리

# Latches & Flip-Flops

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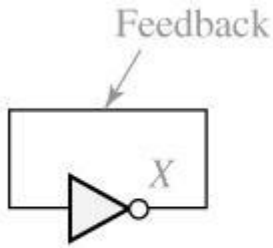
# Types of Logic Circuits

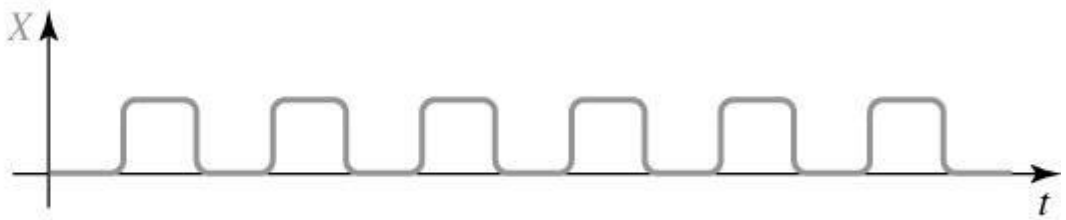
- **Combinational circuits**
  - **The output only depends on the current inputs.**
  - **Memoryless**
- **Sequential circuits**
  - **The output depends on the present input combination and the past sequence of inputs.**
  - **It remembers/stores something of past input history.**
  - **Memory**
    - **Latches** (no clock):
      - state change by inputs
    - **Flip-Flops** (clock for synchronization):
      - state change by the clock
  - **Feedback loop**
    - Outputs are connected to the inputs.

# Feedback Loop

To construct a circuit has memory, introduce *feedback* to circuit

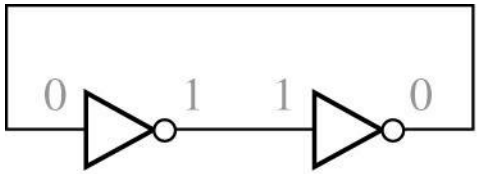



(a) Inverter with feedback

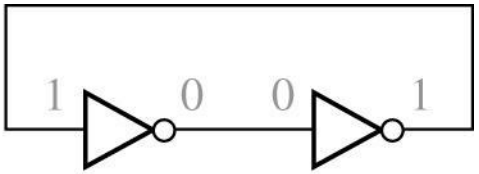



(b) Oscillation at inverter output

➔ Unstable state: oscillation rate depends on the delay of inverter




(a)

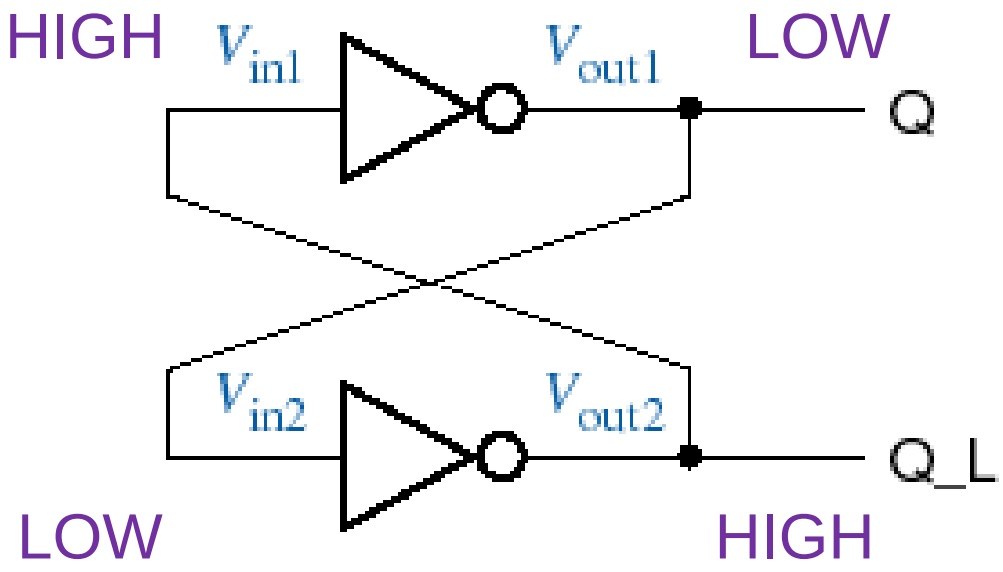



(b)

➔ Stable state

# Bistable element

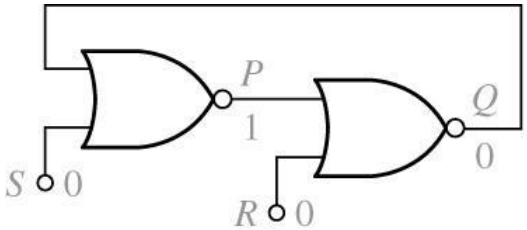
- **The simplest sequential circuit**
- **Two states**
  - **One state variable: Q**



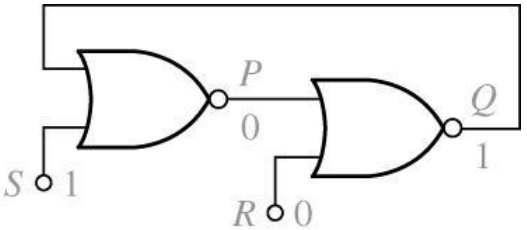
# Set-Reset (SR) Latch

Memory by the past inputs

1) S=R=0 (stable Q=0) → 2) S=1, R=0 (set to Q=1)



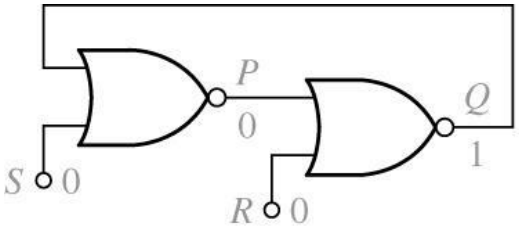

(a)



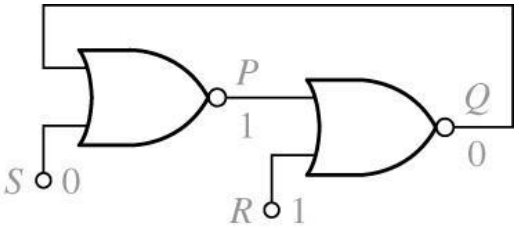

(b)

S changes to 1, **set Q=1**

3) S=R=0 (stable Q=1) → 4) S=0, R=1 (reset to Q=0)




(a)

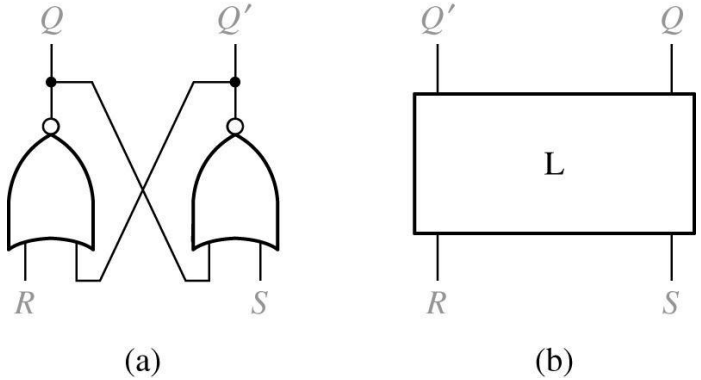



(b)

R changes to 1, **reset Q=0**

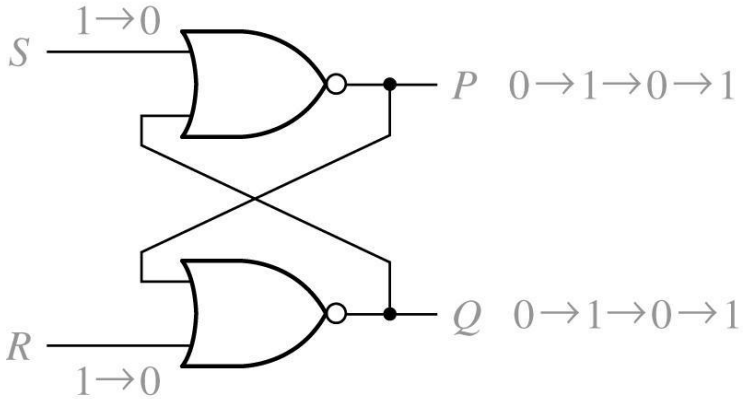
# Cross-coupled structure

- S-R Latch(cross-coupled structure) where S=R=1 is not allowed




Latch outputs should be complement, Q and Q'

- Improper S-R Latch Operation (S=R=1; prohibited)

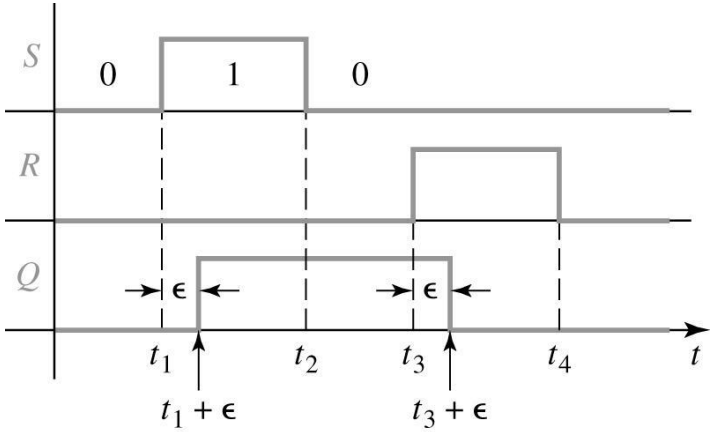



S=R=1, P=Q=0, not complement

S=R=0, P and Q are oscillated.

# Timing diagram

- Timing Diagram for SR Latch: state is changed after propagation delay




- S-R Latch Operation

| $S(t)$ | $R(t)$ | $Q(t)$ | $Q(t+\varepsilon)$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | - |
| 1 | 1 | 1 | - |

State change table
Present state -> Next state

Inputs not allowed

# Next State (or Characteristic) Equation

- Which conditions change the latch state into 1?

| $R(t)\,Q(t)$ \ $S(t)$ | 0 | 1 |
|---|---|---|
| 00 | 0 | 1 |
| 01 | 1 | 1 |
| 11 | 0 | X |
| 10 | 0 | X |

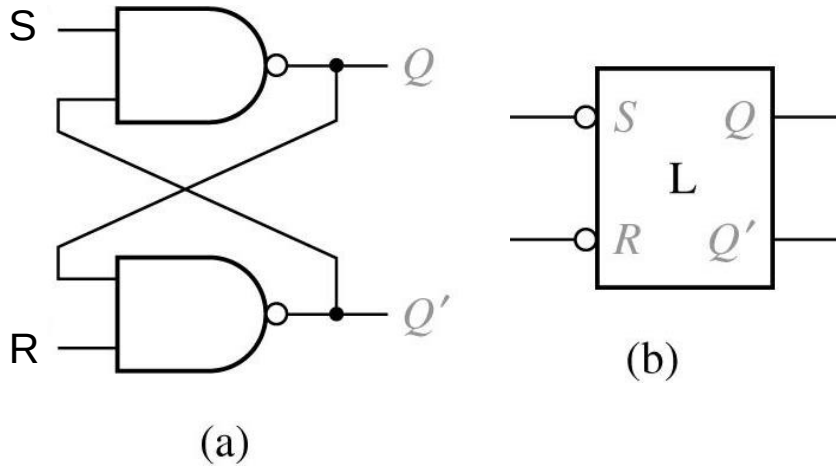
$$Q(t+\epsilon) = S(t) + R'(t)\,Q(t)$$

$$\boxed{Q^{+} = S + R'Q \qquad (SR = 0)}$$

Next-state equation

# $\overline{S}\,\overline{R}$ Latch

- NAND gate SR Latch

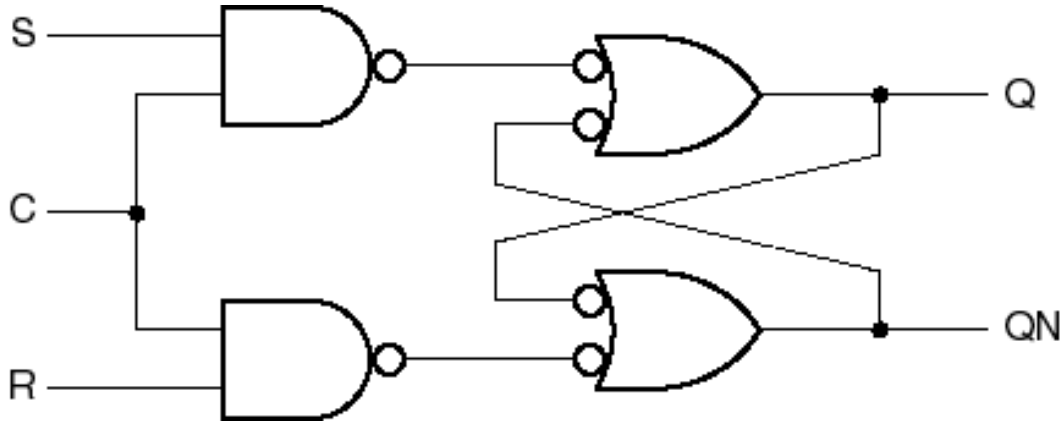



| $S$ | $R$ | $Q$ | $Q^+$ |
|---|---|---|---|
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | - |
| 0 | 0 | 1 | - |

S=0, Q changes to 1
R=0, Q changes to 0

S=R=0 is not allowed

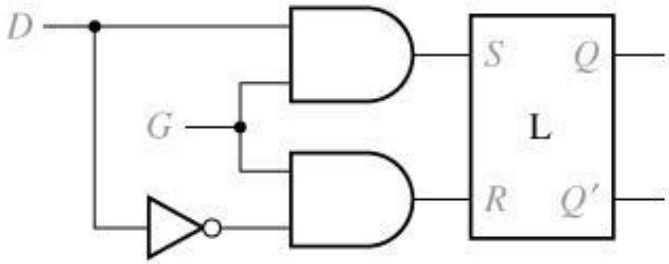
# SR Latch with Enable

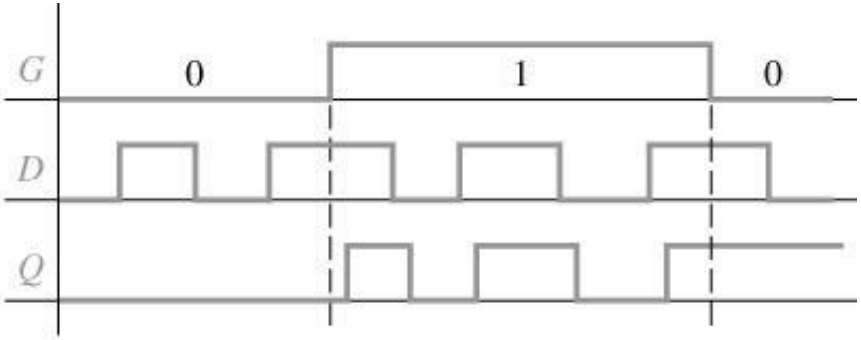



| S | R | C | Q | QN | |
|---|---|---|---|---|---|
| 0 | 0 | 1 | last Q | last QN | |
| 0 | 1 | 1 | 0 | 1 | |
| 1 | 0 | 1 | 1 | 0 | |
| 1 | 1 | 1 | 1 | 1 | → undefined |
| x | x | 0 | last Q | last QN | |

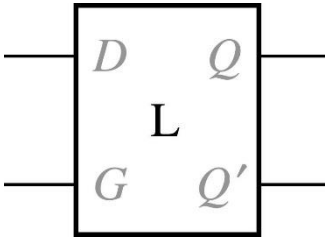
# Gated D Latch

- Q follows data D. (D=1 ->Q=1,  D=0->Q=0)




- Symbol and Truth Table for Gated Latch




| $G$ | $D$ | $Q$ | $Q^+$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

| $Q$ \ $GD$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 0 |

$$Q^+ = G'Q + GD$$